

Exercises (with Solutions) · Chapter 5

1. Problem

A researcher fits a logistic regression model to predict a binary outcome variable $Y \in \{0, 1\}$ using a single ($p = 1$) predictor variable X .

The values of the parameter estimates are $\hat{\beta}_0 = -2.4$ and $\hat{\beta}_1 = 0.8$.

- (a) Write down the fitted logistic regression model for $\hat{P}(Y = 1|X = x)$.
- (b) Compute the value x of the predictor such that $\hat{P}(Y = 1|X = x) = \hat{P}(Y = 0|X = x)$.

Solution

- (a) The fitted logistic regression model for $\hat{P}(Y = 1|X = x)$ is given by

$$\begin{aligned}\hat{P}(Y = 1|X = x) &= \frac{1}{1 + \exp(-(\hat{\beta}_0 + \hat{\beta}_1 x))} \\ &= \frac{1}{1 + \exp(-(-2.4 + 0.8 \cdot x))}.\end{aligned}$$

- (b) To compute the value x of the predictor such that $\hat{P}(Y = 1|X = x) = \hat{P}(Y = 0|X = x)$, observe that

$$\hat{P}(Y = 1|X = x) = \hat{P}(Y = 0|X = x) = 0.5$$

holds true. Thus, we have to solve the equation

$$\begin{aligned}\hat{P}(Y = 1|X = x) &= 0.5 \\ \Leftrightarrow \frac{1}{1 + \exp(-(\hat{\beta}_0 + \hat{\beta}_1 x))} &= 0.5 \\ \Leftrightarrow \exp(-(\hat{\beta}_0 + \hat{\beta}_1 x)) &= 1 \\ \Leftrightarrow -(\hat{\beta}_0 + \hat{\beta}_1 x) &= \ln(1) \\ \Leftrightarrow -\hat{\beta}_0 - \hat{\beta}_1 x &= 0 \\ \Leftrightarrow \frac{-\hat{\beta}_0}{\hat{\beta}_1} &= x \\ \Leftrightarrow \frac{2.4}{0.8} &= x \\ \Leftrightarrow 3 &= x\end{aligned}$$

2. Problem

Consider the following classifier f for a binary outcome variable $Y \in \{0, 1\}$ and a single ($p = 1$) predictor variable X ,

$$f(x_0) = \begin{cases} 1 & \text{if } P(Y = 1|X = x_0) \geq 0.5 \\ 0 & \text{if } P(Y = 1|X = x_0) < 0.5 \end{cases}$$

Let the true (usually unknown) conditional probability function be given by

$$P(Y = 1|X = x_0) = \begin{cases} 1 & \text{if } x_0 \geq 0.6 \\ 0 & \text{if } x_0 < 0.6 \end{cases}$$

You have a training data set of size $n_{\text{Train}} = 5$ generated according to the model above with $X_i = i/n_{\text{Train}}$:

$$\begin{aligned} & \{(X_1, Y_1), (X_2, Y_2), (X_3, Y_3), (X_4, Y_4), (X_5, Y_5)\} \\ &= \{(0.2, 0), (0.4, 0), (0.6, 1), (0.8, 1), (1, 1)\}. \end{aligned}$$

You estimate $P(Y = 1|X = x)$ using the KNN estimator

$$\hat{P}_K(Y = 1|X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(Y_i = 1),$$

applied to the training data set above. Given your estimation $\hat{P}_K(Y = 1|X = x_0)$ you construct the classifier $\hat{f}_K$ as above by replacing the true conditional probability by its estimate $\hat{P}_K(Y = 1|X = x_0)$.

- (a) What is the value of $\hat{P}_K(Y = 1|X = 0.4)$ for $K = 1$ and $K = 5$?
- (b) What is the training error rate for $K = 1$ and $K = 5$?

Solution

- (a) The value of $\hat{P}_K(Y = 1|X = 0.4)$ for $K = 1$ is given by

$$\hat{P}_{K=1}(Y = 1|X = 0.4) = \frac{1}{1} \sum_{i \in \mathcal{N}_0} I(Y_i = 1),$$

where $\mathcal{N}_0 = 2$ such that

$$\hat{P}_{K=1}(Y = 1|X = 0.4) = I(Y_2 = 1) = 0.$$

The value of $\hat{P}_K(Y = 1|X = 0.4)$ for $K = 5$ is given by

$$\hat{P}_{K=5}(Y = 1|X = 0.4) = \frac{1}{5} \sum_{i \in \mathcal{N}_0} I(Y_i = 1),$$

where $\mathcal{N}_0 = \{1, 2, 3, 4, 5\}$ such that

$$\begin{aligned} \hat{P}_{K=5}(Y = 1|X = 0.4) &= \frac{1}{5} (I(Y_1 = 1) + I(Y_2 = 1) + I(Y_3 = 1) + I(Y_4 = 1) + I(Y_5 = 1)) \\ &= \frac{1}{5} (I(0 = 1) + I(0 = 1) + I(1 = 1) + I(1 = 1) + I(1 = 1)) \\ &= \frac{1}{5} (0 + 0 + 1 + 1 + 1) \\ &= \frac{3}{5} \\ &= 0.6 \end{aligned}$$

- (b) The training error rate for $K = 1$ is given by

$$\begin{aligned} & \frac{1}{n_{\text{Train}}} \sum_{i=1}^{n_{\text{Train}}} I(Y_i \neq \hat{Y}_i), \quad \text{where} \quad \hat{Y}_i = \hat{f}_{K=1}(X_i) \\ & \frac{1}{5} (I(0 \neq \hat{f}_{K=1}(0.2)) + I(0 \neq \hat{f}_{K=1}(0.4)) + I(1 \neq \hat{f}_{K=1}(0.6)) + I(1 \neq \hat{f}_{K=1}(0.8)) + I(1 \neq \hat{f}_{K=1}(1.0))) \\ & \frac{1}{5} (I(0 \neq 0) + I(0 \neq 0) + I(1 \neq 1) + I(1 \neq 1) + I(1 \neq 1)) \\ & \frac{1}{5} (0 + 0 + 0 + 0 + 0) = 0, \end{aligned}$$

since

$$\begin{aligned}
 \hat{P}_{K=1}(Y = 1|X = 0.2) &= 0 < 0.5 \Rightarrow \hat{f}_{K=1}(0.2) = 0 \\
 \hat{P}_{K=1}(Y = 1|X = 0.4) &= 0 < 0.5 \Rightarrow \hat{f}_{K=1}(0.4) = 0 \\
 \hat{P}_{K=1}(Y = 1|X = 0.6) &= 1 \geq 0.5 \Rightarrow \hat{f}_{K=1}(0.6) = 1 \\
 \hat{P}_{K=1}(Y = 1|X = 0.8) &= 1 \geq 0.5 \Rightarrow \hat{f}_{K=1}(0.8) = 1 \\
 \hat{P}_{K=1}(Y = 1|X = 1.0) &= 1 \geq 0.5 \Rightarrow \hat{f}_{K=1}(1.0) = 1.
 \end{aligned}$$

The training error rate for $K = 5$ is given by

$$\begin{aligned}
 &\frac{1}{n_{\text{Train}}} \sum_{i=1}^{n_{\text{Train}}} I(Y_i \neq \hat{Y}_i), \quad \text{where } \hat{Y}_i = \hat{f}_{K=5}(X_i) \\
 &\frac{1}{5} \left(I(0 \neq \hat{f}_{K=5}(0.2)) + I(0 \neq \hat{f}_{K=5}(0.4)) + I(1 \neq \hat{f}_{K=5}(0.6)) + I(1 \neq \hat{f}_{K=5}(0.8)) + I(1 \neq \hat{f}_{K=5}(1.0)) \right) \\
 &\frac{1}{5} (I(0 \neq 1) + I(0 \neq 1) + I(1 \neq 1) + I(1 \neq 1) + I(1 \neq 1)) \\
 &\frac{1}{5} (1 + 1 + 0 + 0 + 0) = 0.4,
 \end{aligned}$$

since

$$\begin{aligned}
 \hat{P}_{K=5}(Y = 1|X = 0.2) &= 0.6 \geq 0.5 \Rightarrow \hat{f}_{K=5}(0.2) = 1 \\
 \hat{P}_{K=5}(Y = 1|X = 0.4) &= 0.6 \geq 0.5 \Rightarrow \hat{f}_{K=5}(0.4) = 1 \\
 \hat{P}_{K=5}(Y = 1|X = 0.6) &= 0.6 \geq 0.5 \Rightarrow \hat{f}_{K=5}(0.6) = 1 \\
 \hat{P}_{K=5}(Y = 1|X = 0.8) &= 0.6 \geq 0.5 \Rightarrow \hat{f}_{K=5}(0.8) = 1 \\
 \hat{P}_{K=5}(Y = 1|X = 1.0) &= 0.6 \geq 0.5 \Rightarrow \hat{f}_{K=5}(1.0) = 1.
 \end{aligned}$$